A Graph Reinforcement Learning Framework for Routing in a Hybrid Fiber and Free-Space QKD Network

Author Names Withheld for Draft

***Abstract*—We study simulator-based next-hop routing for a modeled trusted-node quantum key distribution network with fiber and free-space optical links. A graph neural network trained with proximal policy optimization (GNN-PPO) observes normalized link and key-pool state and selects from a shared physical-feasibility mask. We evaluate five independently trained GNN-PPO and multilayer-perceptron PPO policies on six ordered city pairs, six season/time conditions, and three shared environment replicates (108 scenarios per checkpoint), against five routing baselines. Across training seeds, GNN-PPO completed 46.1% of held-out routes (nested 95% bootstrap interval 30.9–58.5%), below MLP-PPO at 79.8% (71.3–87.4%) and the BFS-hop and tuned state-aware weighted-shortest-path baselines at 100%. GNN-PPO exceeded distance-only Dijkstra (28.7%) but showed substantial training-seed variability and longer successful routes. A four-level fiber-outage stress sweep preserved this ranking. These results establish a reproducible comparison and identify a gap between the graph policy and simpler alternatives; they do not support a claim of GNN superiority. The topology and link quantities are simulator proxies, not measured routes or delivered secret-key throughput.**

***Index Terms*—Quantum key distribution, trusted-node networks, routing, graph neural networks, attention, proximal policy optimization, free-space optical communication.**

1. INTRODUCTION

Trusted-node quantum key distribution (QKD) networks extend point-to-point key generation by relaying keys across intermediate nodes. A route determines which links must be available and which stored key resources are consumed, so network-level key delivery depends on more than geometric distance [1]–[4]. Routing therefore connects physical channel conditions to resource availability and end-to-end service feasibility.

Conventional shortest-path methods optimize a fixed edge cost or hop count. In QKD networks, link error, usable key-rate state, stored key pools, and time-varying availability can change which next hop is feasible and desirable [5], [6]. A controller that sees these quantities must choose among feasible alternatives while the state evolves over the episode. This paper asks: *Can graph-structured reinforcement learning exploit heterogeneous physical and resource state for constrained next-hop routing in a hybrid fiber/FSO trusted-node QKD network?*

The question is particularly relevant to hybrid fiber and free-space optical (FSO) links. Fiber attenuation varies predictably with span length and device parameters, whereas an FSO link also depends on atmospheric attenuation, turbulence, pointing, and availability. Under changing scenario state, neighboring links can have different quality and feasibility even when their topological positions are unchanged. A useful routing formulation must expose this state to the decision rule.

Prior QKD-routing studies address related but distinct decision problems. Sharma *et al.* formulate routing and resource assignment for QKD-secured optical lightpath requests, jointly considering route and optical-resource allocation under traffic load [7]. Johann *et al.* study routing in meshed trusted-node networks with key-store state and network-level route selection [6]. Learning-based routing has also been studied in quantum networks, including PPO-based entanglement routing; the use of reinforcement learning itself is therefore not the contribution claimed here [8], [9]. We instead examine graph-structured sequential next-hop decisions where fiber/FSO physical state and normalized resource state enter the policy observation. The task is route selection in a single-episode simulator, not traffic-level routing, wavelength/time-slot assignment, or demand blocking.

We model a scenario-dependent network as a graph with node and link observations. Invalid candidate actions are excluded by an explicit feasibility mask; a three-layer graph neural network (GNN) encodes graph context, and a proximal policy optimization (PPO) actor selects a next hop among the remaining candidates. PPO supplies the optimization procedure, while the research question concerns whether this graph representation and state coupling produce useful choices among feasible alternatives.

The contributions are fourfold: (1) a hybrid fiber/FSO trusted-node simulation framework coupling scenario-dependent channel and normalized key-pool state to routing observations; (2) a constrained GNN-PPO next-hop formulation operating on graph state; (3) a reproducible evaluation protocol with classical comparators and archived route-level evidence; and (4) controlled policy-sensitivity and exact reward-oracle analyses on fixed-context inputs and tractable graphs. The current results show that BFS remains a strong comparator; they do not support a claim of learned-policy superiority over hop-minimizing routing.

Section II reviews QKD links, trusted-node networking, and related routing methods. Section III describes the simulator and experimental protocol. Section IV formulates the constrained routing policy and reward. Section V reports the available paired evaluation, reward audit, and small-graph

This manuscript is a working draft based on the QKD routing simulator maintained by the authors. Author affiliations, funding, and correspondence details are to be completed.

oracle. Section VI interprets these findings and their limits, and Section VII concludes.

2. BACKGROUND AND RELATED WORK

A. QKD and Trusted-Node Networking

QKD protocols combine a quantum channel with authenticated classical communication. BB84 uses nonorthogonal signal states to allow the communicating parties to estimate disturbance and distill key material [10]. Practical transmitters commonly use weak coherent pulses rather than ideal single-photon sources. Decoy-state methods use pulses with different intensities to estimate single-photon contributions and counter photon-number-splitting attacks [11], [12]. Finite-key operation must include statistical fluctuations, error-correction leakage, and security parameters. Asymptotic rate formulas are useful for link-level screening, but they do not by themselves certify a finite block. Entropic uncertainty and composable security analyses provide established foundations for finite-size bounds [13]–[15]. For STN chains, the parties and parity announcements couple information across links. A finite-key bound for such a chain is therefore protocol-specific and cannot be replaced by summing independent per-link asymptotic rates [16].

In a conventional trusted-node chain, neighboring QKD modules establish link keys. The network forwards end-to-end key material through trusted relays, commonly using one-time-pad operations. For a path with link key rates R_i, the instantaneous end-to-end service rate is limited by the slowest required resource, and practical delivery also depends on stored key inventory and demand. Key pools therefore couple routing decisions over time [4], [6].

B. Fiber and FSO QKD Links

Long fiber spans motivate intermediate trusted nodes. FSO links can provide candidate bypasses where a direct fiber span is unavailable or degraded. Their channel quality depends on atmospheric attenuation, turbulence, pointing, and availability. Terrestrial FSO design methods identify these as separate budget terms and require local environmental information for credible planning [17], [18]. The simulator models time-of-day and seasonal scenario variables, but its current FSO distributions are assumptions rather than fits to site-specific Indian weather traces.

The routing quantity used in this work is a normalized simulator rate proxy, not an experimentally validated secret-key throughput. The physical model maps attenuation, detector efficiency, and dark-count settings to detection and QBER estimates; its optional decoy-state finite-key engineering mode is not a composable security proof. FSO weather and turbulence distributions are synthetic scenarios rather than site-calibrated meteorological traces.

C. QKD Routing and Resource Management

Classical shortest-path methods provide reproducible baselines but optimize a fixed edge cost or hop count [19]. QKD network research has therefore studied key-store use, request blocking, and route/resource assignment. Sharma *et al.* use deep reinforcement learning to jointly solve routing and resource assignment for QKD lightpath requests in optical networks; their action includes optical resource allocation and their metrics include blocking and resource utilization under traffic [7]. Johann *et al.* formulate meshed trusted-node route selection around key-store state and network-level requests [6]. These formulations address multi-request service and resource assignment, while the present simulator isolates constrained next-hop selection for a single route and does not model wavelength/time-slot assignment or demand blocking. Other learning-based work studies entanglement routing or mixed trusted-node/repeater settings [8], [9]; these differ in network resources and route semantics.

D. Graph Reinforcement Learning

Sequential decision methods can represent state-dependent link availability; dynamic programming provides a classical foundation for sequential optimization [20]. PPO is a clipped policy-gradient method, and generalized advantage estimation reduces variance in its return estimates [21], [22]. The partially observed formulation follows standard POMDP and reinforcement-learning foundations [23], [24]. Graph neural networks encode neighborhoods by message passing, for example

$$h_v^{(\ell+1)} = \phi\Big(h_v^{(\ell)}, \mathrm{AGG}_{u\in\mathcal{N}(v)}\, \psi(h_v^{(\ell)}, h_u^{(\ell)}, e_{uv})\Big), \quad (1)$$

with graph convolution, neighborhood aggregation, and attention providing established operators [25]–[27]. This representation is relevant when candidate actions are neighboring nodes and link state must be combined with route context. Whether it improves decisions over a matched non-graph policy remains an empirical question.

3. SIMULATOR ARCHITECTURE AND EXPERIMENTAL SETUP

A. Architecture

The simulator couples topology and link profiles to a time-stepped environment, routing policy, and evaluation layer. At each step, scenario variables update fiber/FSO link states; the environment forms graph and candidate-edge observations, applies a shared feasibility mask, and updates route and normalized key-pool state after the selected hop. Fig. 1 summarizes this physical-to-decision coupling. The controller is a centralized simulator abstraction, not a deployed QKD management interface.

B. Simulation Setup

1. Topology and Corridor Data

The topology contains seven city hubs (Delhi, Jaipur, Mumbai, Bangalore, Hyderabad, Kolkata, and Chennai), 2,370 modeled nodes, and 2,653 links: 273 fiber links and 2,380 candidate FSO links. OpenStreetMap driving routes provide corridor geometry for the modeled fiber paths; this is a planning proxy, not verified deployed fiber infrastructure. FSO line of sight and terrain clearance are not surveyed. The topology is archived with the research data and uses OpenStreetMap data under its applicable attribution and database terms [28].

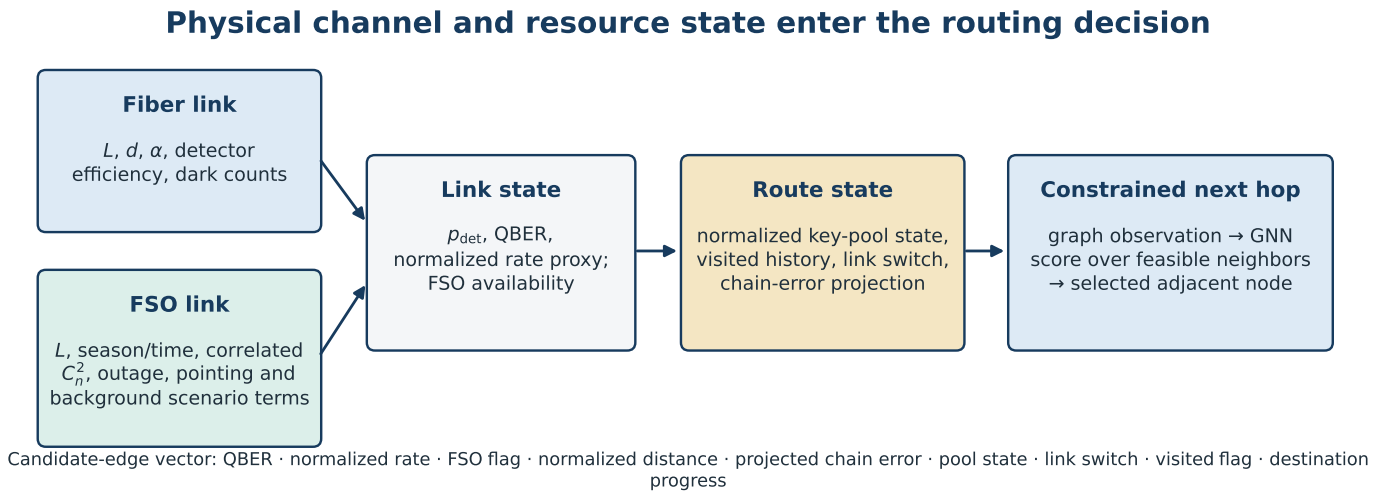


Fig. 1. Physical channel and normalized resource state flow into the constrained routing decision. Fiber and FSO model inputs produce link state; route state joins that information in candidate-edge features before the masked GNN-PPO action is selected.

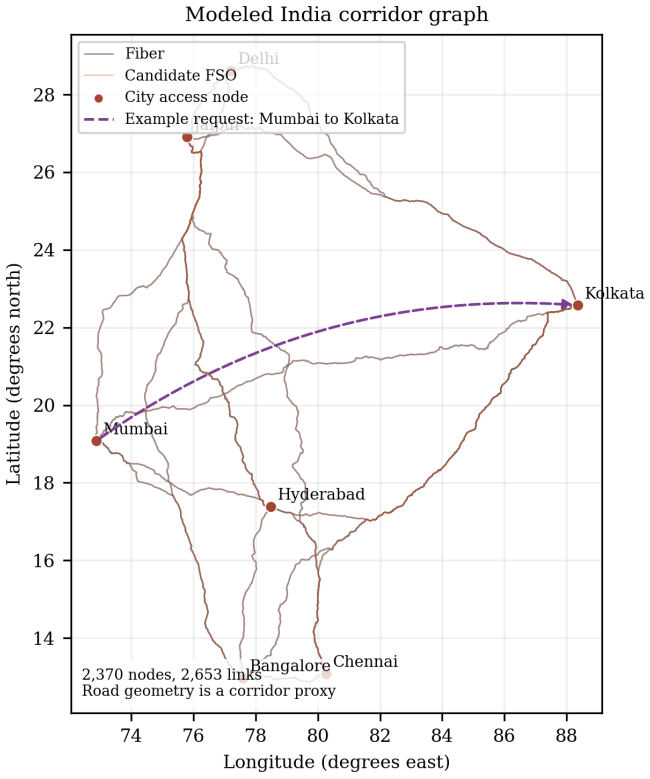


Fig. 2. Modeled corridor graph with fiber links and candidate FSO links. The dashed Mumbai-to-Kolkata line marks an example request, not a physical link. Road geometry is a corridor proxy; FSO line of sight is assumed.

TABLE I
REPRESENTATIVE LINK-MODEL PARAMETERS AND EVIDENCE CLASS

Quantity	Configured value	Evidence class
Fiber wavelength	1550 nm	device profile
Fiber attenuation	0.20 dB/km	inherited target
Fiber detector efficiency	0.93	measured device value
Fiber dark-count bound	$< 70\ \mathrm{s}^{-1}$	measured upper bound
FSO wavelength	785 nm	scenario assumption
FSO detector efficiency	0.60	scenario assumption
FSO dark counts	$100\ \mathrm{s}^{-1}$	scenario assumption
FSO temporal correlation	0.85	uncalibrated assumption

2. *Link and Noise Models*

Fiber loss is represented using wavelength-dependent attenuation and excess loss. Detector efficiency, dark-count bounds, and a visibility assumption contribute to detection and QBER proxies. The selected fiber device profile is based on published/vendor-reported ID Quantique ID281 values at 1550 nm. The FSO channel is a separate assumed 785-nm scenario with its own efficiency and dark-count settings. These are not a single matched commercial optical terminal profile.

FSO turbulence is represented with a correlated latent log-C_n^2 process. The environment samples availability and channel conditions by time step; FSO corridor outages are held fixed for the episode to avoid a bypass disappearing between its relay segments. These choices create repeatable stress scenarios. A detailed parameter ledger and provenance categories are supplied in `PHYSICS_EVIDENCE.md`.

The implemented FSO model uses a correlated log-normal turbulence process with atmospheric, diffraction, pointing, and background terms. Detector and channel noise enter through the QBER and normalized rate calculations. The optional finite-key estimator is an engineering model, not an independently verified composable proof.

The default rate mode is an asymptotic, normalized proxy of the form

$$r_{\mathrm{proxy}} = p_{\mathrm{det}} \left[1 - 2h_2(Q)\right], \tag{2}$$

TABLE II
OBSERVATION FEATURES FOR EACH CURRENT NODE AND CANDIDATE EDGE

Feature group	Components
Node (5)	City indicator; full-TN indicator; STN indicator; destination-relative distance; normalized degree.
Candidate edge (9)	QBER; normalized rate proxy; FSO indicator; normalized distance; projected chain error; normalized key-pool state; link-type switch; visited-node flag; destination-distance progress.
Message-passing edge (4)	QBER; normalized rate proxy; FSO indicator; normalized distance.
Feasibility	Adjacent node; QBER below threshold; positive rate proxy; projected chain error below threshold.

where p_{det} is modeled detection probability, Q is modeled QBER, and h_2 is binary entropy. An opt-in decoy-state mode estimates finite-key contributions and uncertainty; its engineering estimate is not a composable security proof.

3. Environment and Protected Conditions

The observation uses relative node features and candidate-edge features; absolute coordinates and node identifiers are excluded in the evaluated configuration. Candidate actions are adjacent nodes only. Let $\mathcal{N}(v_t)$ be the neighbors of the current node and define $m_{t,u} = 1$ only if edge (v_t, u) has QBER below $Q_{\max}$, positive normalized rate proxy, and projected chain error below $Q_{\max}$. The feasible action set is

$$\mathcal{A}_{\mathrm{feasible}}(s_t) = \{u \in \mathcal{N}(v_t) : m_{t,u} = 1\}, \tag{3}$$

with $Q_{\max} = 0.11$. This mask enforces feasibility; the policy's utility is measured by its selection among the remaining feasible neighbors.

4. Reward Definition

Feasibility and utility are distinct: the mask in (3) removes invalid actions, while the shaped reward ranks feasible moves. For the defence profile used by the archived experiments, the per-step return is the sum of link, route, and terminal terms:

$$\begin{aligned} r_t = {} & w_p R_{p,t} + w_q R_{q,t} + w_r R_{r,t} + w_k (R^{\mathrm{level}}_{k,t} + R^{\mathrm{delta}}_{k,t}) \\ & - C_{\ell,t} - C_{e,t} - C_{c,t} + C_{\mathrm{step},t} + C_{\mathrm{revisit},t} + C_{\mathrm{switch},t} \\ & + C_{\mathrm{backtrack},t} + C_{\mathrm{arrival},t} + C_{\mathrm{security},t} + C_{\mathrm{depletion},t}. \end{aligned} \tag{4}$$

The progress, QBER-margin, rate, and pool terms are signed by destination-distance progress; link costs and terminal penalties are applied as implemented. Table III gives the defence-profile weights and fixed protection values used in the current reward audit.

5. Training and Evaluation Setup

The archive contains earlier single-checkpoint transfer and reward-audit runs. To replace those mixed-stage results, the completion protocol was frozen before new runs: five independent GNN-PPO and five MLP-PPO policies, matched BC/PPO budgets, endpoint exclusion from both training stages, a monsoon-only validation suite, and a paired test suite over held-out endpoint pairs and normal/summer/winter conditions. The protocol and resolved per-seed configurations are supplied in the supplementary package. This separation keeps checkpoint selection on validation data and reserves the held-out test suite for final reporting.

TABLE III
REWARD TERMS FOR THE DEFENCE PROFILE

Term	Weight/value	Per-step contribution
Progress	0.2	Destination-distance reduction, normalized by max edge distance.
QBER margin	0.8	Feasible projected-chain margin, signed by progress.
Rate proxy	1.2	Distance-prorated normalized link-rate utility, signed by progress.
Pool level / change	0.4 each	Current normalized pool and its change, signed by progress.
Latency / energy	0.4 each	Distance-based costs with configured normalization constants.
Congestion	0.5	Degree cost above two incident choices at the departure node.
Step / revisit	-0.05 / -0.6	Fixed step and revisit costs.
Switch / backtrack	-0.2 / -0.5	Link-type switch and non-progress cost.
Arrival	$+4.0$	Destination terminal bonus.
Security / depletion	-30 / -60	Terminal penalties for chain-error breach and pool depletion.

TABLE IV
FROZEN PROTOCOL FOR THE FINAL MATCHED ARCHITECTURE AND HOLDOUT STUDY

Setting	Value
PPO endpoint and training season	Delhi–Chennai; monsoon, 22:00
BC demonstrations	Random pairs excluding test pairs and reversals
Independent training seeds	20261002–20261006 per architecture
Matched architectures	GNN-PPO and pointwise MLP-PPO
Training budget	15 BC epochs; 50 PPO epochs; 144 steps/epoch
Validation	Delhi–Chennai; monsoon; seeds 11001–11004
Held-out endpoints	Six ordered pairs; reversals excluded from BC
Held-out environment	Normal, summer, winter; 02:00 and 22:00
Paired test suite	108 scenarios; unique seeds across cells; all policies
QBER / outage / episode limit	0.11 / 0.01 / 144 steps

Test comparators are Random, BFS-hop, Dijkstra-km, Max-SKR, and a receding-horizon weighted shortest path whose normalized distance, QBER, rate-proxy, and key-pool costs are tuned only on validation scenarios. The test pairs and seeds are common to every method. The full edge-cost expression and selection grid are fixed in the protocol file before testing.

4. ROUTING POLICY AND LEARNING OBJECTIVE

A. Graph State and Constrained Routing

At time t, let $G_t = (V, E, X_t)$ denote the modeled graph and its scenario-dependent link state, where X_t contains channel and normalized pool values. The environment state s_t additionally contains current and destination nodes, route history, chain-error state, and time. The policy receives observation o_t and selects an adjacent node $a_t \in \mathcal{A}_{\mathrm{feasible}}(s_t)$ as defined in (3). Transitions update the route, key-pool level, and channel scenario; future turbulence and outage draws are not exposed as a forecast to the policy.

An episode succeeds upon reaching the destination. It terminates as a failure after a projected security-bound breach or modeled pool depletion, and is truncated at the configured

step limit if neither terminal condition has occurred. If no feasible neighbor remains, the policy has no selectable action and the episode is recorded as a no-valid-action failure by the evaluator. These conditions define route completion within this simulator and are not a cryptographic security certification.

The objective is to find a policy $\pi_\theta(a_t \mid o_t)$ that maximizes expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right], \tag{5}$$

subject to the action constraint in (3). The reward in (4) is a training signal; evaluation reports route success, successful-route hop count, invalid-action/fallback counts, and condition and seed variation.

1. Graph Policy and Action Mask

For each node v, the message-passing stack computes an embedding h_v from node features and graph connectivity. The encoder uses QBER and positive rate proxy to filter invalid full-graph edges; the message function itself aggregates learned transformations of endpoint embeddings. The actor's attention head then combines current, destination, and candidate-neighbor embeddings with the candidate's nine route-context features. It produces logits $z_{v,u}$ for adjacent candidates. Invalid candidates are assigned negative infinity before softmax:

$$\pi_\theta(u \mid o_t) = \mathrm{softmax}_{u \in \mathcal{A}(o_t)}(z_{v_t,u}). \tag{6}$$

Invalid links contribute no message in either direction, and the candidate mask additionally enforces projected chain error. This separates the environment's feasibility filter from the learned preference: the policy chooses among feasible alternatives.

In the optional attention variant, candidate embeddings are combined using learned compatibility scores, following neighborhood attention formulations [27]. Attention is treated as an architectural choice. Its value must be established by an ablation with the same training budget and seed distribution. The current results do not isolate a causal benefit from attention.

2. PPO Training

The policy is updated by maximizing the clipped surrogate objective

$$L^{\mathrm{clip}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, \mathrm{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{7}$$

where $\rho_t(\theta)$ is the new-to-old action probability ratio and $\hat{A}_t$ is the estimated advantage. PPO is combined with a value loss and entropy regularization. Hyperparameters are resolved from a versioned YAML file and included with the experiment artifact. The settings for the transfer-evaluation checkpoint and paired episode seeds are archived in the supplementary package.

B. Classical Comparators

BFS minimizes hop count on the modeled topology, and Dijkstra minimizes fiber/FSO distance. Random routing samples uniformly from the feasible action set; Max-SKR greedily selects its highest-rate candidate. The state-aware comparator recomputes current edge weights at each decision and selects the next hop minimizing its edge cost plus the minimum remaining path cost:

$$\begin{aligned} c_e = {} & \frac{d_e}{d_{\max}} + w_Q \frac{Q_e}{Q_{\max}} \\ & + w_R[-\log(\max(r_e, 10^{-6}))] + w_K(1 - K_e), \end{aligned} \tag{8}$$

where r_e is normalized rate proxy and K_e is normalized pool state. The lookahead graph omits links currently failing the local QBER, positive-rate, or nonempty-pool checks; the selected first hop must also pass the full route-dependent action mask. Weights are selected on the fixed validation suite only; the test suite is not used for tuning. This centralized receding-horizon comparator has access to current graph state but no future channel forecast.

5. RESULTS

A. Frozen Multi-Seed Held-Out Comparison

The final evaluation followed the frozen protocol in `supplementary/final_experiment_protocol_20261003.md`. It used six ordered city pairs, six season/time conditions, and three shared environment replicates, giving 108 scenarios per policy/checkpoint. Five GNN-PPO and five matched MLP-PPO checkpoints were each evaluated on all 108 scenarios; five nonlearned policies were evaluated on the same scenario matrix. The held-out comparison therefore contains 1,620 policy-scenario episodes. A separate validation grid selected the Weighted-SP coefficients before testing; the selected tuple was $(w_Q, w_R, w_K) = (1, 0, 0)$, so its resulting edge cost uses distance and current feasibility. Training-seed variability is summarized across the five independently trained checkpoints, and the reported nested bootstrap resamples training seeds and scenario clusters. This is simulator-level evidence for the stated topology and protocol, not field generalization.

Across training seeds, GNN-PPO completed 249/540 routes (46.1%; nested 95% interval 30.9–58.5%), while MLP-PPO completed 431/540 (79.8%; 71.3–87.4%). The five per-seed GNN completion rates ranged from 18.5% to 63.9%; MLP-PPO ranged from 66.7% to 87.0%. BFS-hop and Weighted-SP each completed 108/108 scenarios, whereas Dijkstra-km completed 31/108 (28.7%; scenario-bootstrap interval 20.4–37.0%); Random and Max-SKR completed none. Thus, the measured GNN policy exceeded distance-only Dijkstra by 17.4 percentage points, but trailed MLP-PPO by 33.7 points and the two shortest-path policies by 53.9 points. The strongest result for the GNN is the limited comparison with Dijkstra and the greedy rate rule, not superiority over simpler feasible-path policies. Figures 4 and 5 summarize completion and successful-route length.

Across the five checkpoints, the mean of per-seed successful-route hop averages was 87.8 ± 9.2 for GNN-PPO and 57.8 ± 5.6 for MLP-PPO; BFS-hop and Weighted-SP averaged 15.5 and 15.7 hops, respectively. Random and Max-SKR had no successful routes, so their conditional hop count is undefined. The hop comparison conditions on success and must be read alongside completion rates; it is not an end-to-end

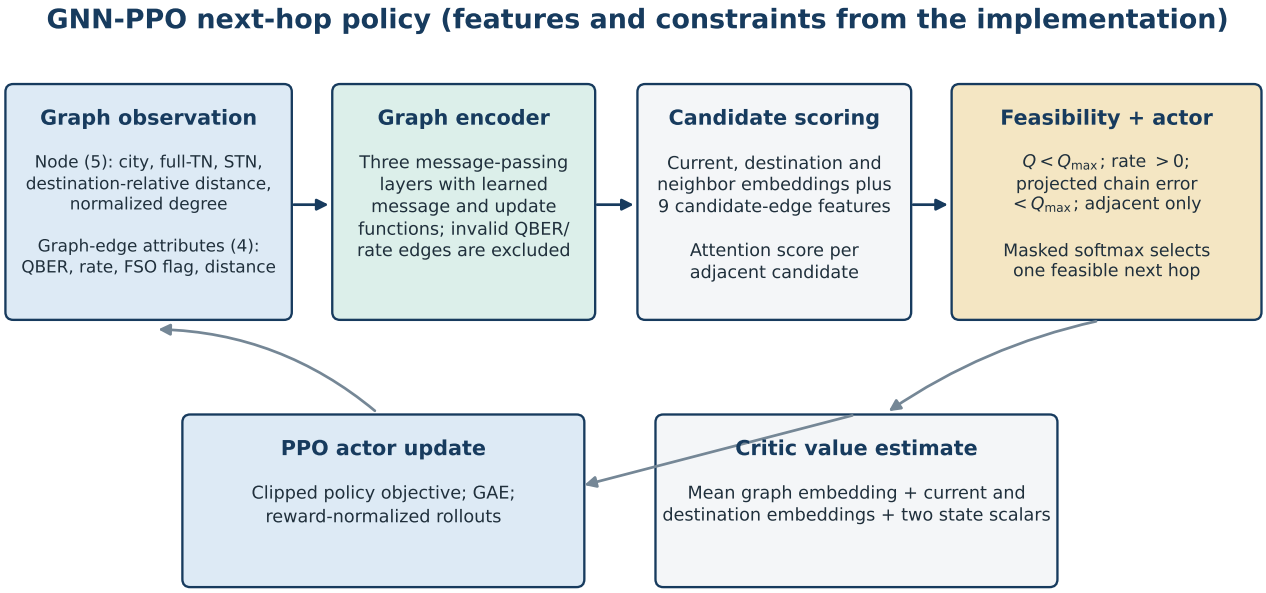


Fig. 3. GNN-PPO policy path with implemented graph inputs, candidate features, feasibility conditions, and actor-critic components. Full-graph edges are filtered by QBER and positive rate before node-only message aggregation; candidate scoring separately consumes the nine route-context features.

efficiency estimate for failed routes. Across the six season/time conditions, the completion gap between the learned policies persisted (Fig. 6). The four-level fiber-outage stress sweep (Fig. 7) likewise showed GNN-PPO between 42.6% and 46.3%, MLP-PPO between 80.6% and 80.9%, and BFS-hop at 100%; Weighted-SP was 99.1% at 10% outage and 100% at the other levels. The small variation with outage does not erase the performance gap, and the sweep does not model correlated or weather-driven outages.

Across the five checkpoints, mean per-decision CPU inference was 6.35 ms (GNN-PPO) and 1.00 ms (MLP-PPO); the means of checkpoint-specific p50, p95, and p99 percentiles were 5.85, 9.61, and 13.33 ms for GNN-PPO and 0.92, 1.48, and 1.86 ms for MLP-PPO. Each checkpoint used 1,000 deterministic measured calls after warm-up, four CPU threads, and disabled encoder caching. These one-host per-decision measurements exclude environment simulation and do not establish deployment latency.

B. Reward-Component Audit

A separate matched reward-component audit evaluated one selected training seed (20261002) for GNN-PPO and MLP-PPO, plus BFS-hop, on the same 108 scenarios per method. Completion was 50/108 (46.3%) for GNN-PPO, 94/108 (87.0%) for MLP-PPO, and 108/108 for BFS-hop. Successful routes averaged 92.56, 58.37, and 15.46 hops, respectively. The component ledger records cumulative shaped rewards along trajectories. GNN-PPO accumulates larger positive QBER-margin, rate-proxy, and pool-level terms than BFS, but also incurs greater step and revisit costs and completes fewer routes. These cumulative terms scale with the visited trajectory and are not independent measures of delivered key service; they do not overturn the completion and hop results. Figure 8 visualizes per-component mean differences against BFS, while the complete episode ledger and summary are included in the experiment archive.

C. Exact Reward Oracle on Small Graphs

To test policy behavior against a tractable optimum, we generated five connected seven-node graphs and evaluated four channel seeds per graph. Link states were frozen within each episode, with a 5% per-edge fiber outage probability. Exhaustive enumeration of feasible simple paths produced an oracle for all 20 cases. The policy checkpoint (SHA-256 prefix `f6e311c2566f`) succeeded in 18/20 cases and matched oracle reward in 11/20 overall and 11/18 successful episodes. Median regret was zero; mean regret was 0.663 among successful episodes and 2.209 when failures were included. This static toy-graph result answers a distinct reward-optimality question and is not an optimality claim for the full dynamic network.

6. DISCUSSION AND LIMITATIONS

A. When Does Graph-Based Routing Help?

The GNN-PPO policy completed more routes than distance-only Dijkstra in the strict matrix, showing a limited benefit over that fixed geometric heuristic under the simulator's changing feasibility state. However, MLP-PPO achieved substantially higher completion with the same PPO objective, and BFS-hop and the validation-tuned Weighted-SP completed nearly every route with far fewer hops. The matched architecture comparison therefore does not support an advantage from graph message passing in the present setup. Seed variability in GNN-PPO is also large relative to its mean; robustness and training stability remain concerns rather than demonstrated strengths.

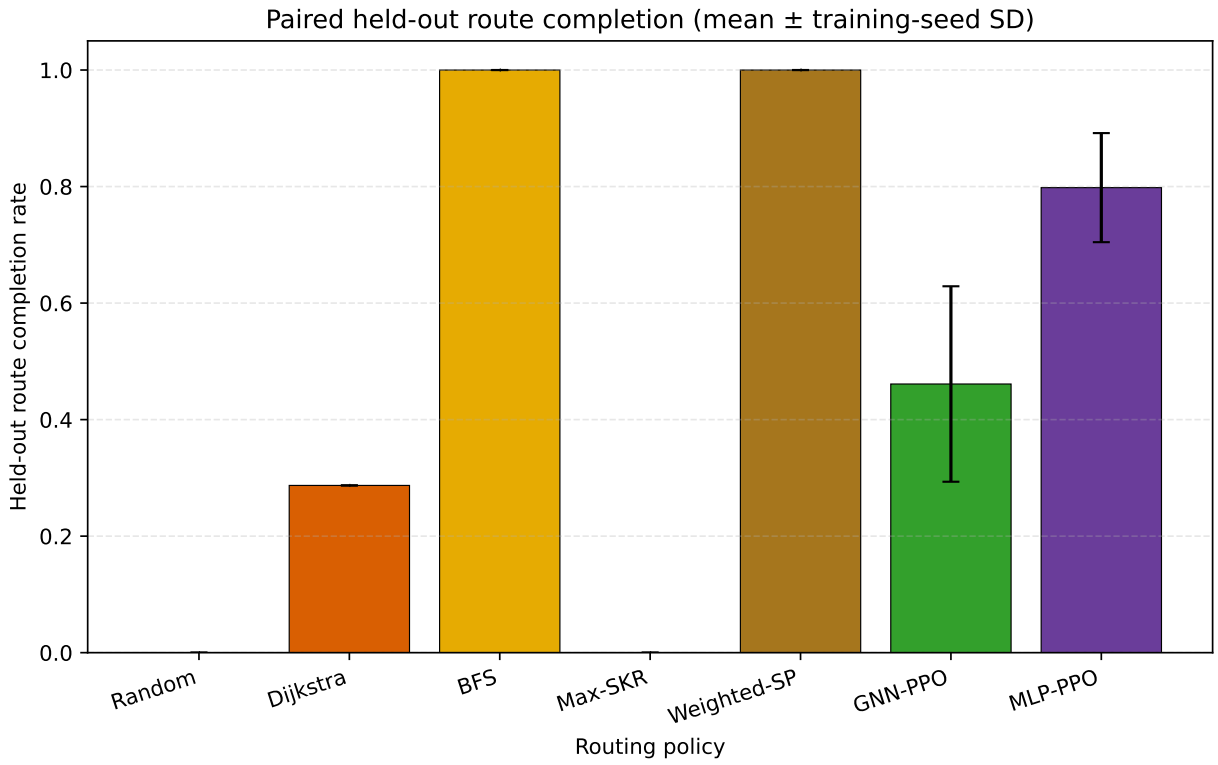


Fig. 4. Strict held-out route-completion rate across the six-pair, six-condition test matrix. Learned-policy error bars show training-seed standard deviation; the nested bootstrap intervals are reported in the text.

B. Why BFS Remains a Strong Baseline

Every method receives the same current-state feasibility mask. BFS therefore chooses among links already filtered by QBER, positive normalized rate, and projected chain error rather than searching the raw graph without physical constraints. In this experiment, that mask and topology make shortest-hop routing highly effective. This is an important interpretation of the benchmark: the learned policy is compared with a strong feasible-path baseline, and the results do not justify replacing it.

C. What State Affects the Policy?

Controlled paired-input counterfactuals hold the route context fixed while varying one candidate feature. Across the tested range, FSO action probability increased from 0.311 to 0.638 as its normalized rate-proxy ratio to fiber rose from 0.25 to 4.0. With QBER below the feasibility threshold, action probability changed only slightly (approximately 0.473 to 0.472); at QBER 0.111 the candidate was masked. In a separate 32-trial turbulence-input sweep, a 100-fold C_n^2 multiplier reduced median FSO rate proxy from 0.202 to 0.183, while mean FSO action probability shifted from 0.749 to 0.740 and FSO remained selected in every trial. The checkpoint is sensitive to proxy-rate contrast and the hard mask, while this particular turbulence scaling had little effect on its local preference. These counterfactual inputs do not establish route-level benefits or calibrated atmospheric response.

D. Model and Network Scope

The archived physical checks give internally consistent outputs for the configured equations: at 80 km and 1550 nm, modeled fiber loss including excess loss is 16.46 dB, detection probability is approximately 0.021 per incident photon, QBER proxy is 0.0105, and normalized rate proxy is 0.0175. The count-rate bound assumes one photon per pulse and is separate from the weak-coherent-pulse finite-key engineering mode. For a 10-km FSO span in the monsoon 22:00 scenario, configured availability is 88% and a 1,000-sample check yielded 88.5%; median normalized rate proxy conditional on availability was 0.2035. These are model checks, not device or field measurements.

Several limits define the present interpretation. The city corridors are OpenStreetMap road proxies, not verified fiber routes; FSO line of sight and weather distributions are uncalibrated. Fiber and FSO parameter sets do not describe one characterized terminal pair. Key-pool state is normalized and single-route: there is no calibrated generation/consumption process, simultaneous demand, or network-wide blocking metric. Management traffic is not counted. The normalized rate proxy is not delivered secret-key throughput, and the optional finite-key estimator is not independently established as composably secure. The frozen five-seed held-out comparison, CPU inference measurements, outage sweep, and reward audit are complete, but remain simulator-only evidence. Matched feature/reward/architecture ablations and empirical calibration are outside the scope of this manuscript revision and are deferred.

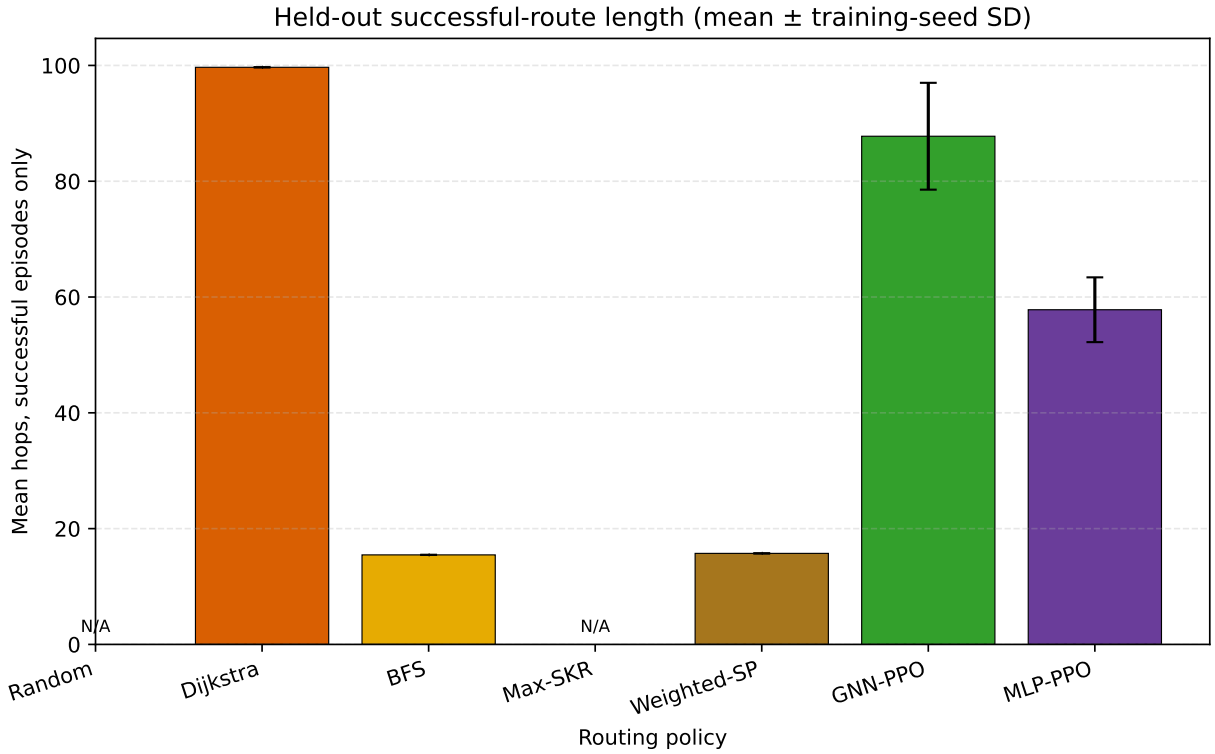


Fig. 5. Mean successful-route length across the strict held-out matrix. Learned-policy bars show the mean of per-seed means with training-seed standard deviation. "N/A" indicates no completed routes, so conditional hop count is undefined.

7. CONCLUSION

This work implemented and evaluated a masked GNN-PPO next-hop policy for a modeled hybrid fiber/FSO trusted-node network. The strict five-seed comparison found 46.1% route completion for GNN-PPO, below matched MLP-PPO (79.8%) and the BFS-hop and tuned Weighted-SP baselines (100% each), while exceeding distance-only Dijkstra (28.7%). The GNN also used substantially more hops on successful routes and had higher inference latency than MLP-PPO. Its measured strength is therefore limited to improving completion over selected weak heuristics; this study does not establish that graph structure improves routing. Reward decomposition, the four-level fiber-outage sweep, and the small-graph oracle provide additional diagnostics but do not measure network throughput or field performance. Future work should target the observed GNN completion and seed-stability gap, then calibrate channels, key pools, and service demand against traceable network data before making deployment claims.

APPENDIX A
SUPPLEMENTARY FIGURES FROM THE PROJECT ARCHIVE

The following figures preserve the additional graphics supplied in the project figure archive. Captions distinguish conceptual diagrams, model-generated quantities, and exploratory comparisons from the final matched held-out experiment. The current implementation diagrams in Figs. 1 and 3 remain the authoritative descriptions of the implemented observation and policy paths.

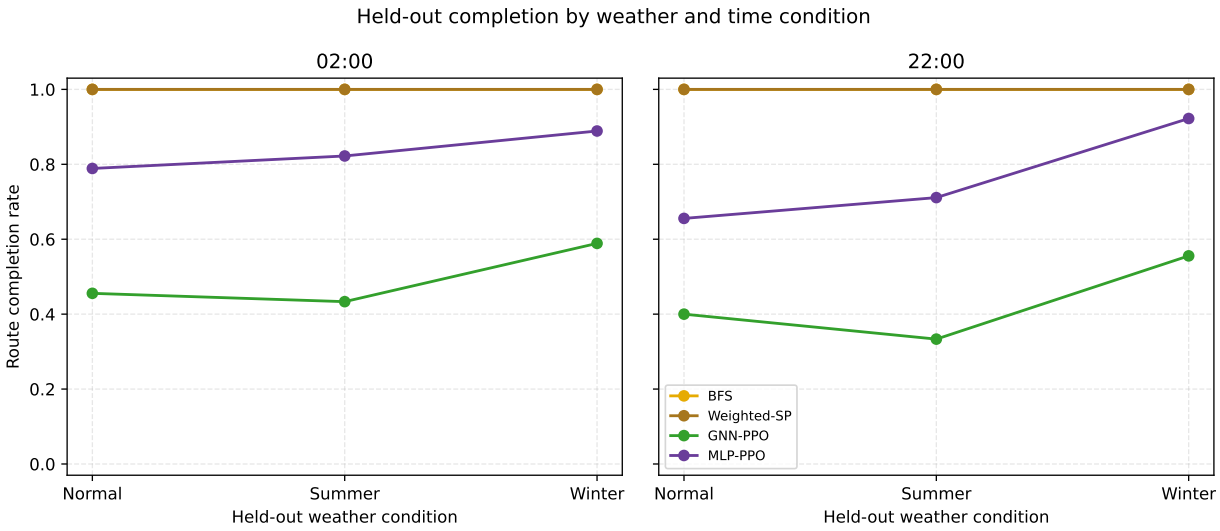


Fig. 6. Held-out route completion across normal, summer, and winter scenarios at 02:00 and 22:00.

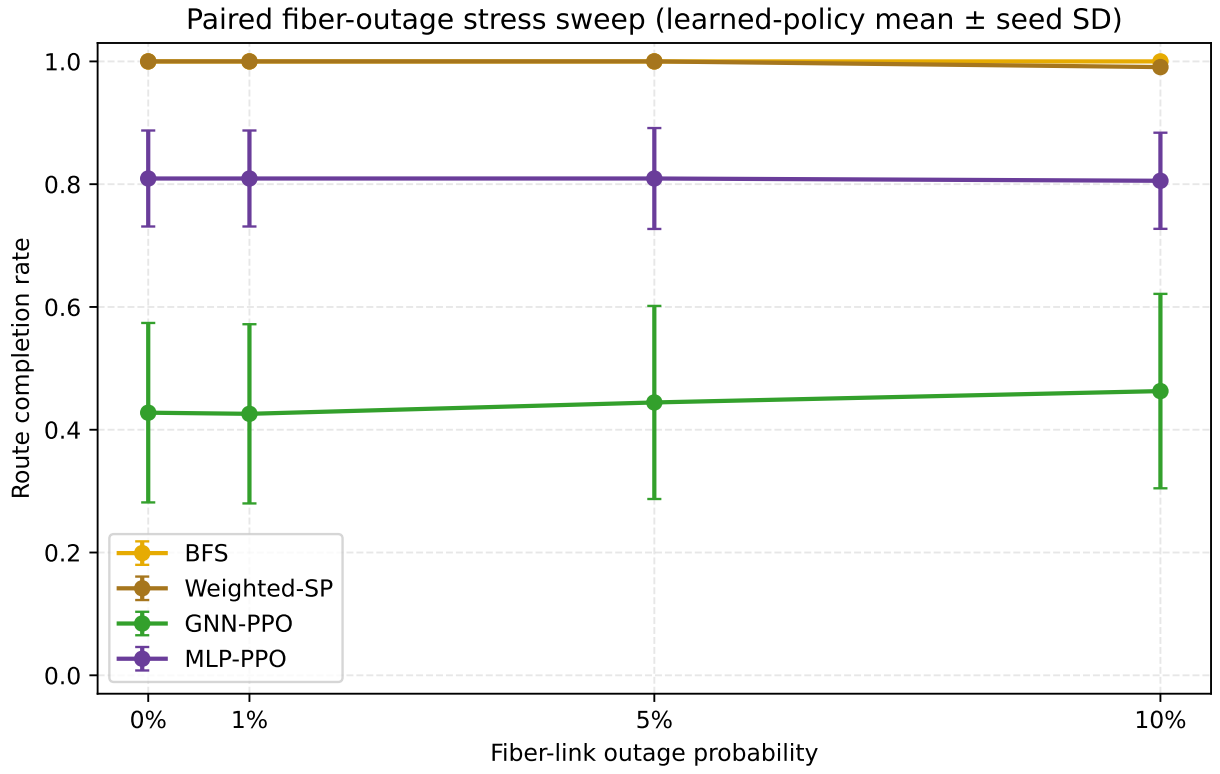


Fig. 7. Fiber-outage stress sweep. Outage levels are per-link simulator probabilities, not calibrated field outage rates; learned-policy error bars show training-seed standard deviation.

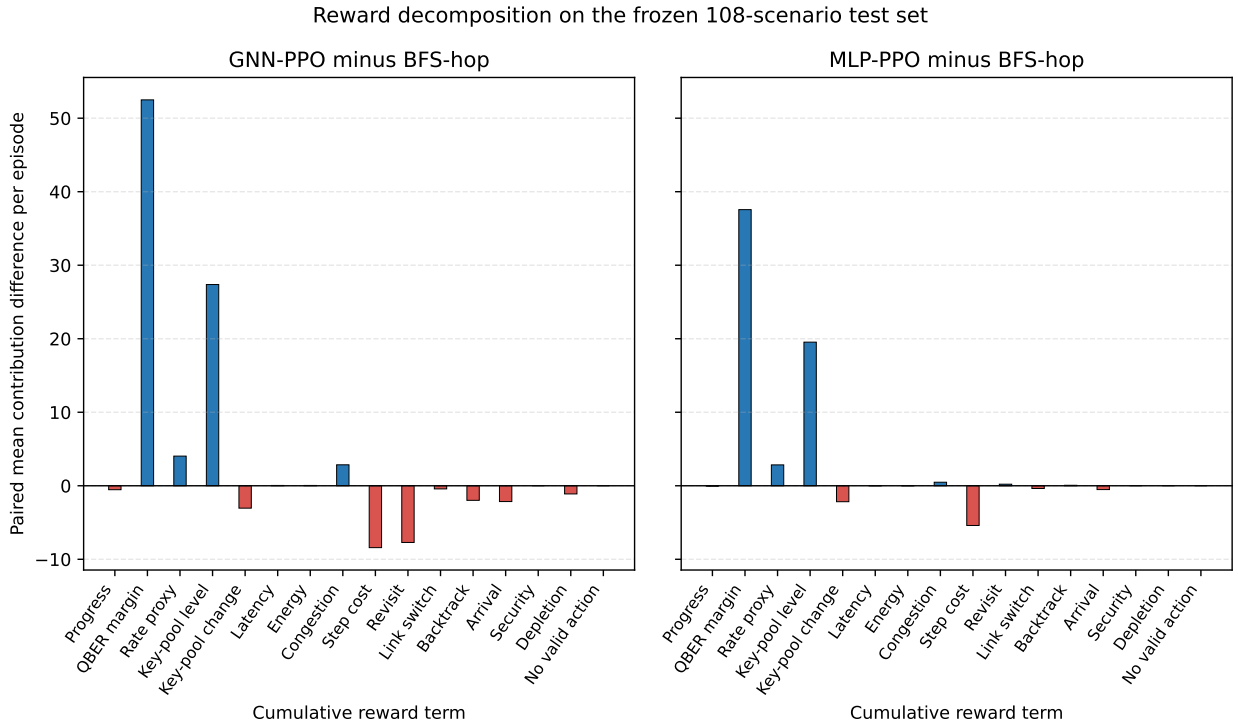


Fig. 8. Cumulative reward-component differences for the selected GNN-PPO and MLP-PPO checkpoints relative to BFS-hop on 108 paired scenarios. Positive values indicate a larger cumulative component contribution; route lengths differ, so the plot is a descriptive reward audit rather than a resource-service comparison.

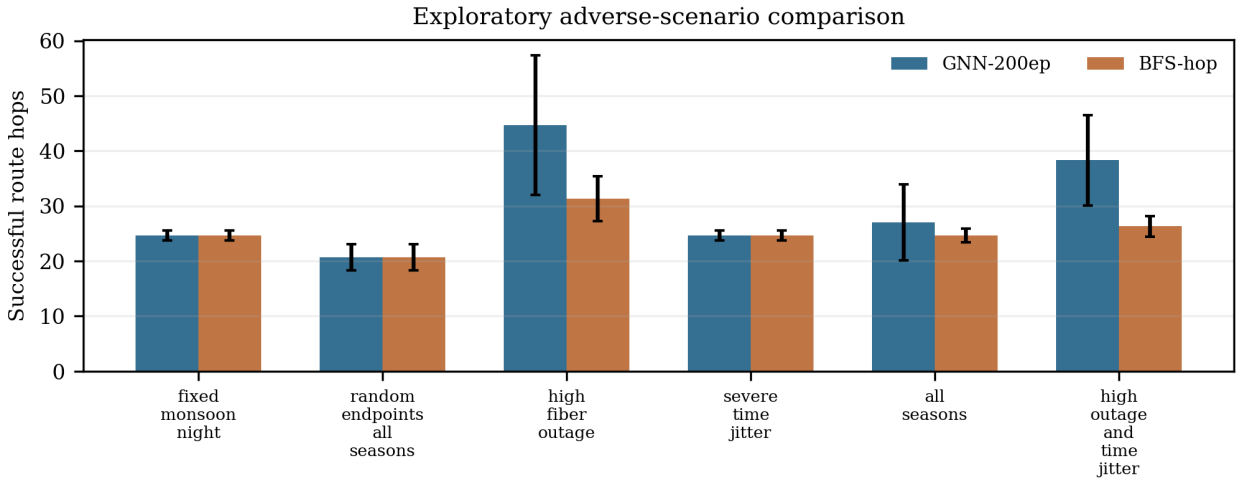


Fig. 9. Exploratory successful-route hop comparison for an earlier GNN-PPO checkpoint and BFS-hop under selected adverse scenarios. This archived diagnostic uses a different run protocol and is not pooled with or treated as evidence from the final matched holdout.

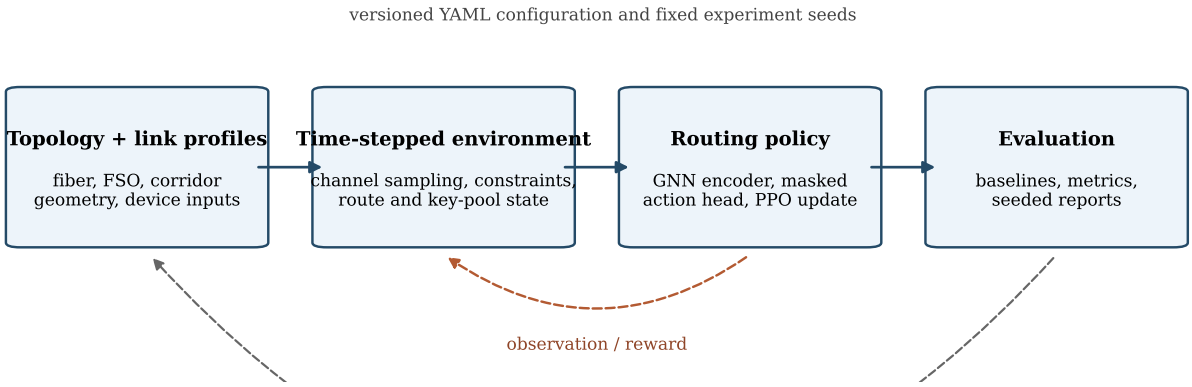


Fig. 10. High-level project workflow showing topology and link profiles, time-stepped environment, routing policy, and evaluation. This is an overview diagram rather than a specification of individual model layers.

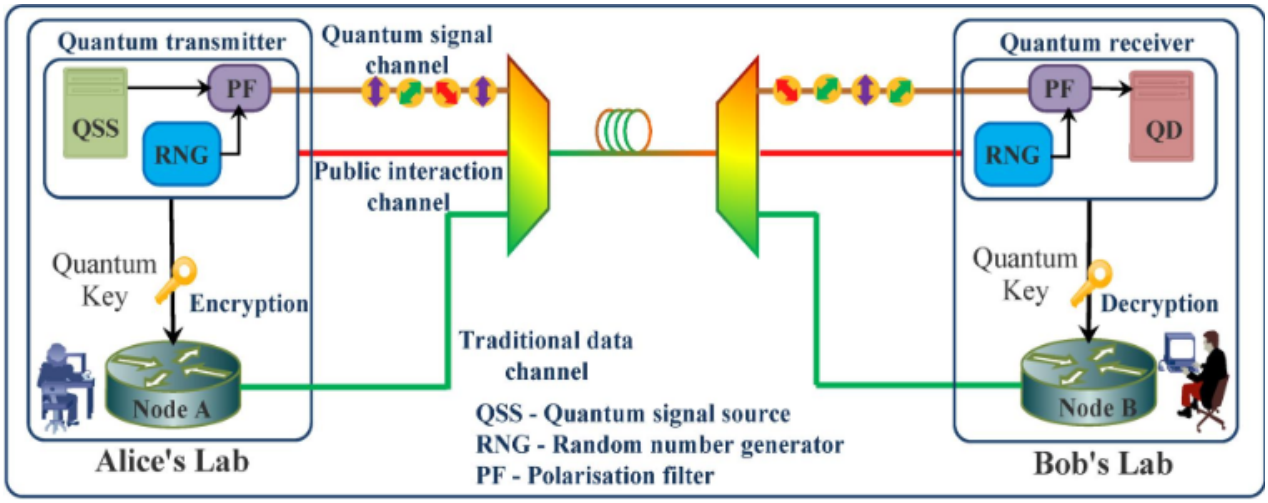


Fig. 11. Conceptual distinction between a quantum signal channel and a public classical interaction channel in a prepare-and-measure QKD link. The drawing is educational context, not a hardware diagram of the simulated network.

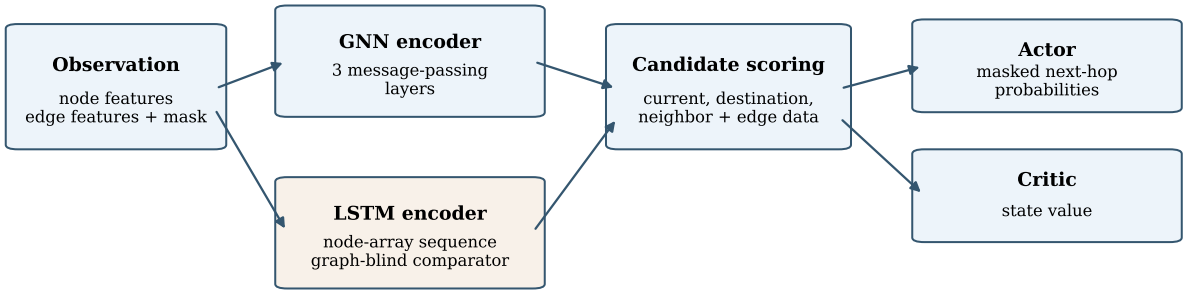


Fig. 12. Earlier conceptual GNN-PPO architecture sketch. It is retained as a project design artifact; the implementation-backed actor–critic and feature flow are specified in Fig. 3.

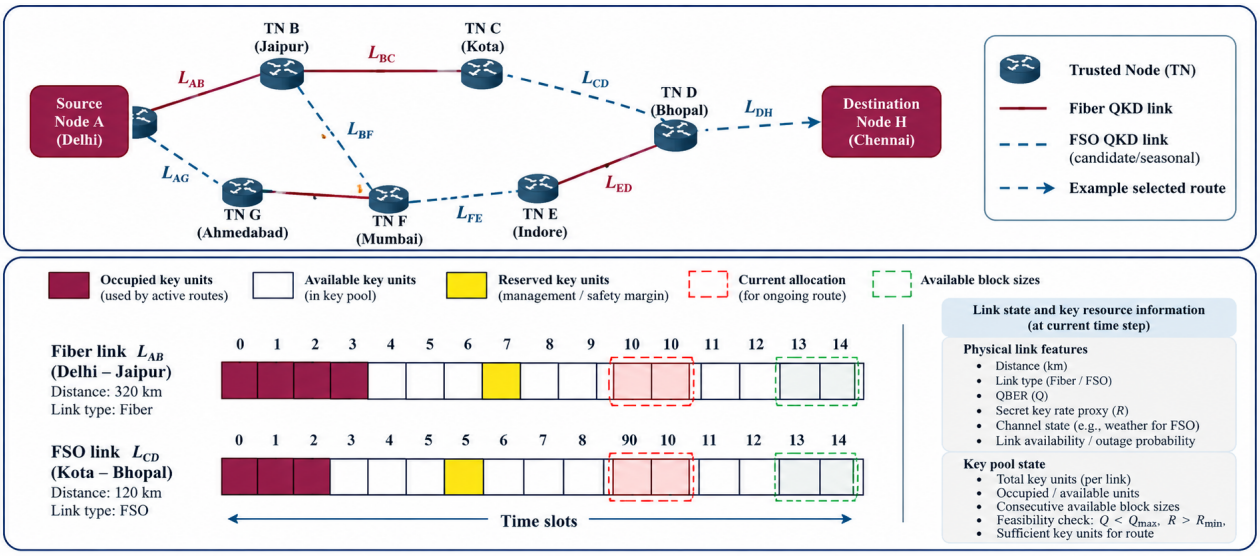


Fig. 13. Conceptual hybrid QKD topology and normalized key-pool visualization. Pool states in the simulator are abstractions, not measured link capacities or a calibrated network-wide key-management process.

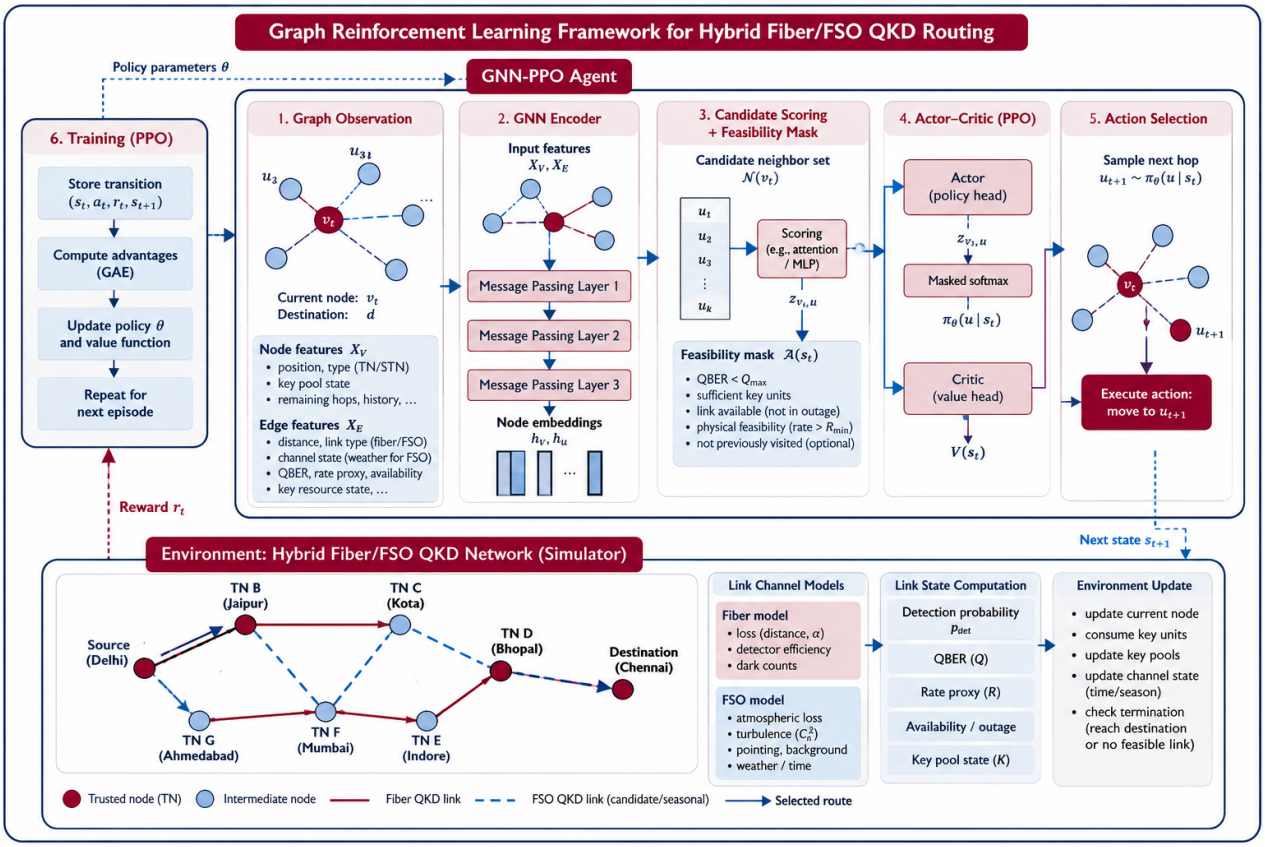


Fig. 14. High-level hybrid-routing and reinforcement-learning workflow from an earlier project design. It is a conceptual overview; implementation details and any differences from the final code are resolved by Fig. 3 and the method description.

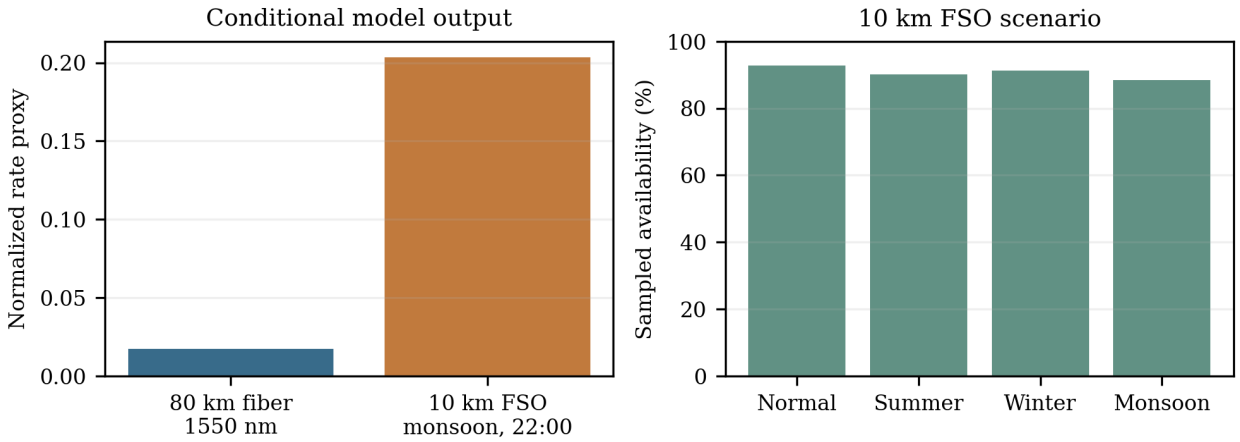


Fig. 15. Simulator-generated normalized rate proxy for representative fiber and FSO links and sampled FSO availability across configured seasons. These conditional model outputs are not field or device measurements.

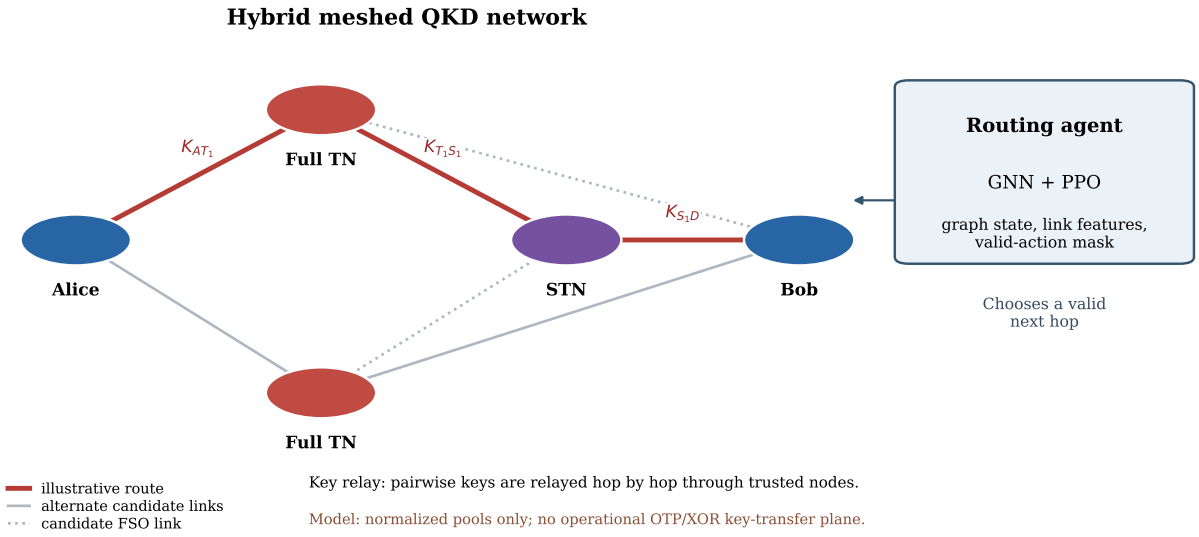


Fig. 16. Conceptual meshed trusted-node QKD architecture with direct and relayed paths. The figure illustrates the routing abstraction and does not assert deployed topology or key-throughput measurements.

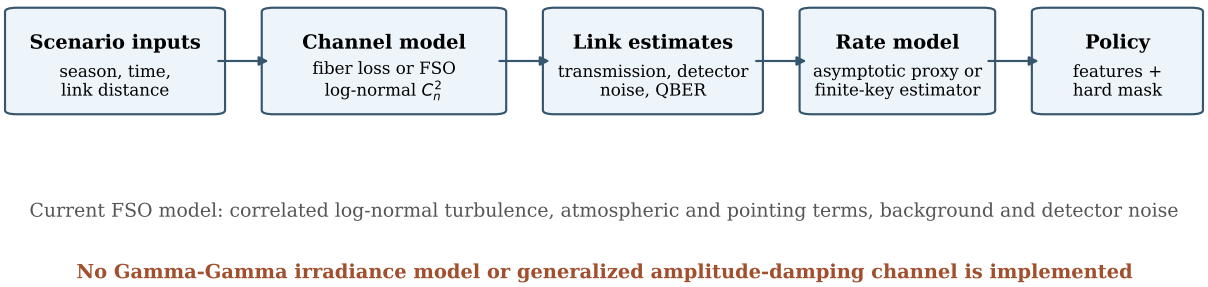


Fig. 17. Conceptual mapping from scenario inputs through channel and rate models to a routing decision. The model quantities and evidence limitations are detailed in Section III and the supplementary evidence ledger.

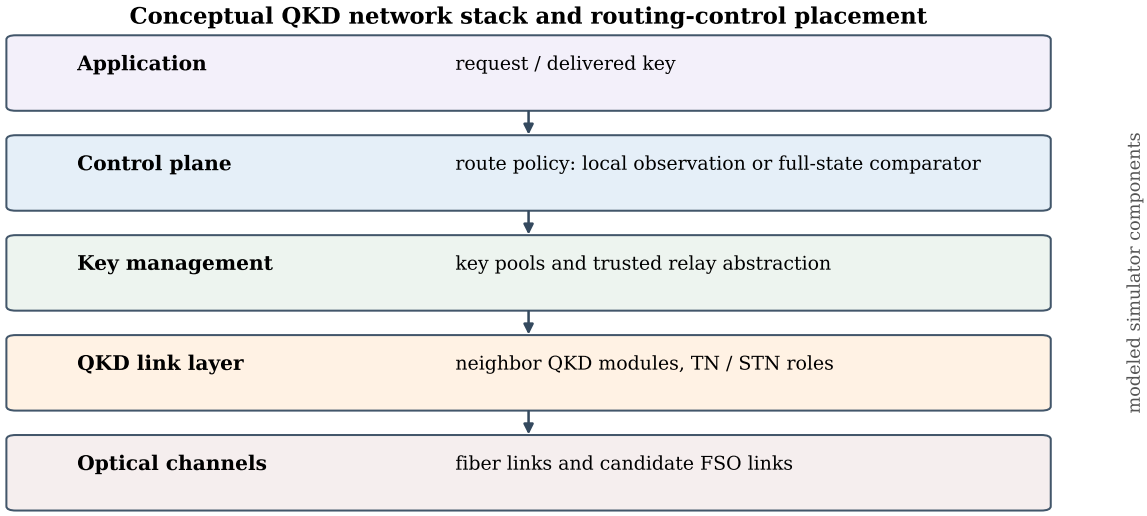


Fig. 18. Conceptual QKD network stack placing routing control, key management, trusted-node relaying, and optical channels in context. It is a functional abstraction rather than a deployed protocol stack.

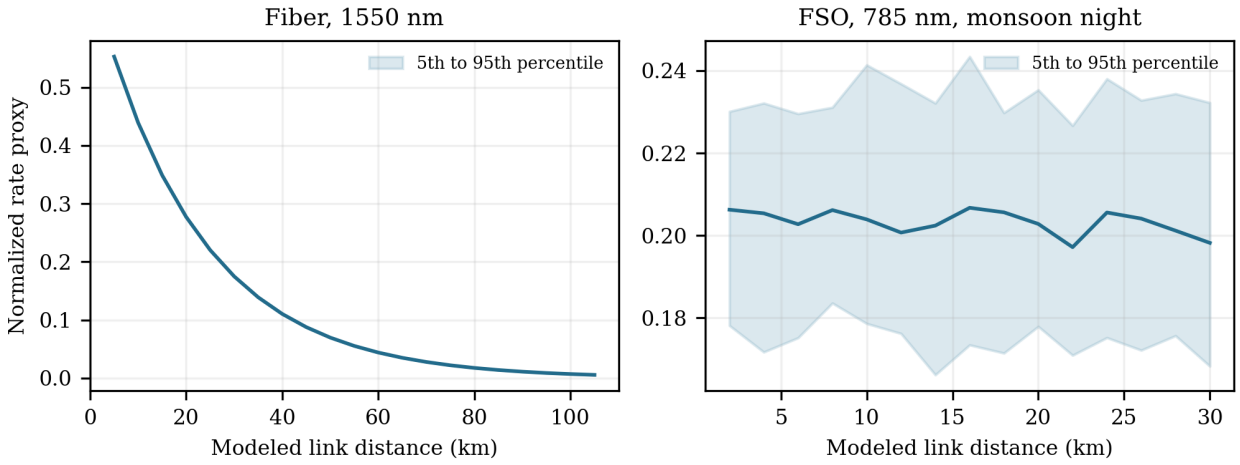


Fig. 19. Configured distance-dependent fiber rate proxy and conditional FSO rate proxy with sampled interval bands. These curves summarize the simulator's assumptions and do not represent calibrated link measurements or secret-key throughput.

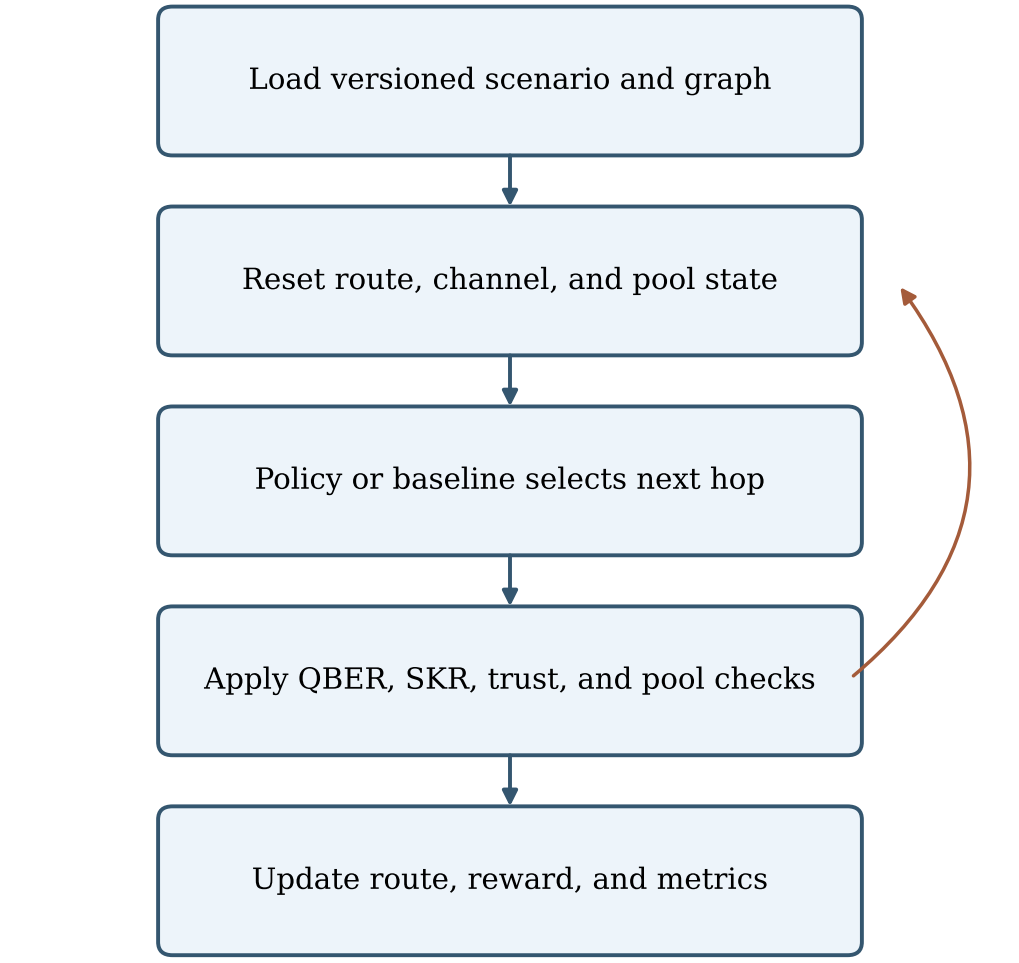


Fig. 20. Episode simulation loop: load the graph and scenario, reset route/channel/pool state, select a next hop, apply feasibility and reward updates, and repeat until termination.

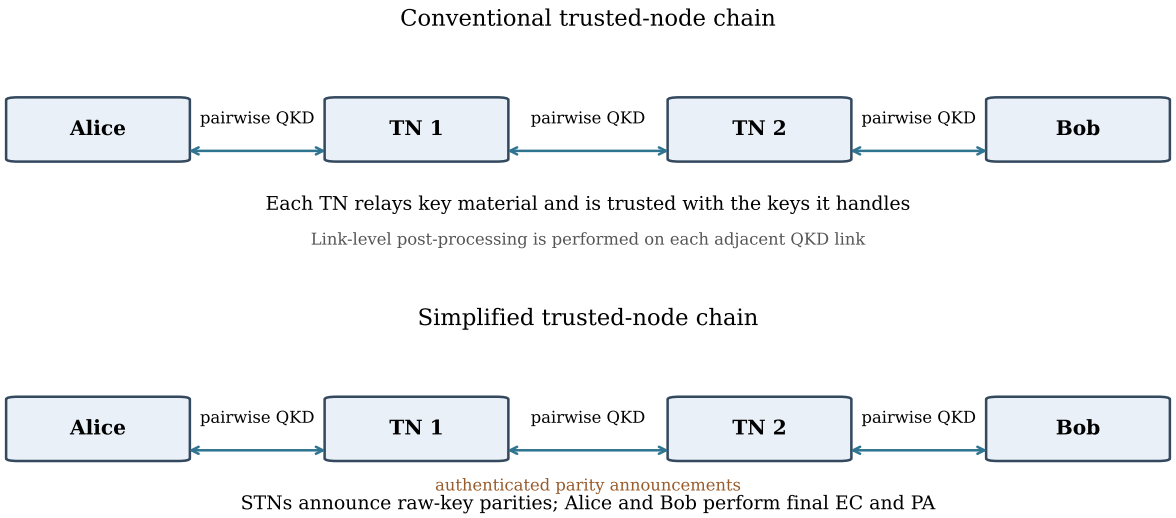


Fig. 21. Conceptual comparison of conventional trusted-node and simplified trusted-relay chain abstractions. The diagram is explanatory and does not define an implemented composable security protocol.

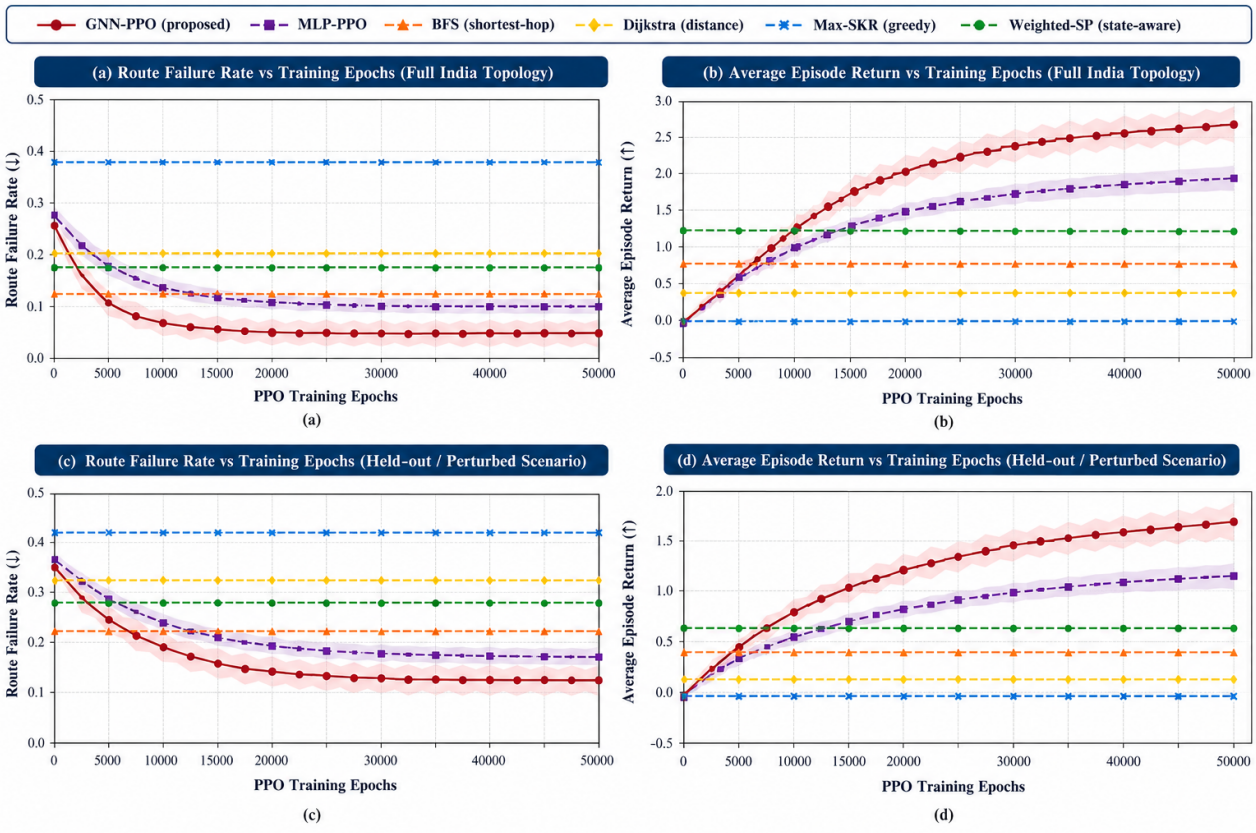


Fig. 22. Illustrative training-curve graphic retained from the project archive. The plotted curves are explicitly illustrative, not logged outcomes from the trained seeds; final training evidence is reported only from archived run histories.

ACKNOWLEDGMENT

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DATA AND CODE AVAILABILITY

The simulator code, archived configurations, scenario outputs, and generated figures are included with the supplementary research package. The road-corridor geometries derive from OpenStreetMap data and are provided under the Open Database License. Publications using the map should credit "Map data from OpenStreetMap contributors" and link to https://www.openstreetmap.org/copyright. The geometries are routing proxies and do not represent verified fiber infrastructure.

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